

Amounts and proportions

Exercise 4 — ISS dataviz, Spring 2026

YOUR NAME HERE

Invalid Date

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Task 1: Session check-in

Three interesting or exciting things:

1. Something
2. Something
3. Something

Three muddy or unclear things:

1. Something
2. Something
3. Something

Task 2: Long Beach Animal Shelter

```
# Load libraries here
library(tidyverse)
```

Data description and cleaning

This data comes from the City of Long Beach Animal Care Services via the {animalshelter} R package (and it was a #TidyTuesday dataset in March 2025).

There are 22 columns, and they're documented here—make sure you consult that page!

The code below loads and cleans the data, mostly by filtering it to include the most common types of animals, intake types, and outcome types. **You don't need to modify anything in this chunk of code—you only need to run it.**

```
# Load the full data
animals_raw <- read_csv("data/animals.csv")

animals <- animals_raw |>
  # Only keep the most common animals, intake types, and outcome types
  filter(animal_type %in% c("bird", "cat", "dog", "wild", "other")) |>
  filter(intake_type %in% c("owner surrender", "stray", "wildlife")) |>
  filter(
    outcome_type %in%
      c("outcome_type", "rescue", "adoption", "euthanasia", "transfer",
        "return to owner", "shelter, neuter, return", "died",
        "community cat", "return to wild habitat")
  ) |>
  # Add columns for the year and month
  mutate(
    intake_year = year(intake_date),
    intake_month = month(intake_date, label = TRUE, abbr = FALSE)
  ) |>
  mutate(
    across(c(animal_type, intake_type, outcome_type),
      \(x) str_to_title(x))
  )
```

Animal type

What are the most common type of animals seen by the shelter?

Recreate data summary

Right now there's a row for each animal visit. We need to condense that down to get counts of different types of variables. We can do this by using `group_by()` and `summarize()`.

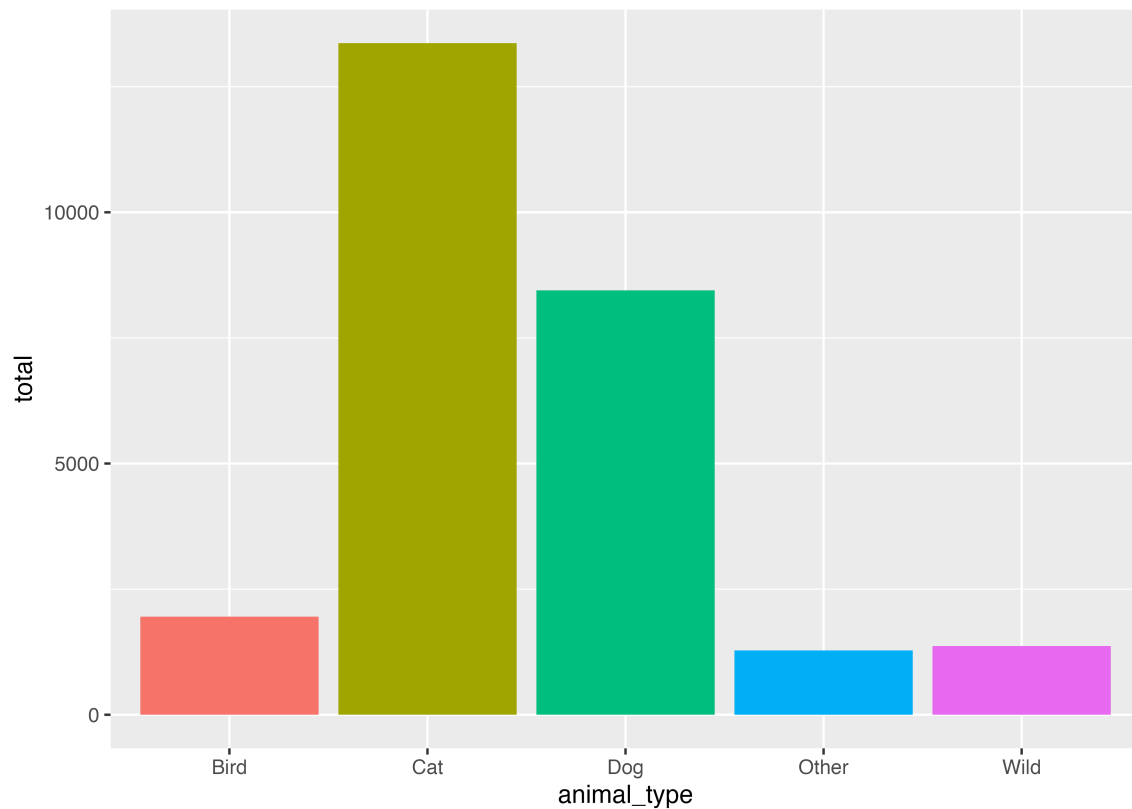
Use `group_by()` and `summarize()` to create a data frame named `animal_counts` that looks like this (hint—in `summarize()`, you'll want to use the `n()` function to calculate the number of rows in each group):

```
# A tibble: 5 × 2
  animal_type total
  <chr>      <int>
1 Bird        1952
2 Cat       13367
3 Dog        8448
4 Other       1280
5 Wild       1366
```

```
# Do stuff here
```

Recreate bar plot

Run the code chunk below to see a plot.



Use `ggplot()` and `geom_col()` in the chunk below to re-create the plot above using the `animal_counts` dataset. (*note*—the bars are filled, but there's no legend)

```
# Do stuff here
```

Outcomes

What are the most common types of outcomes for animals taken to the shelter? What proportion are adopted or rescued?

Recreate data summary

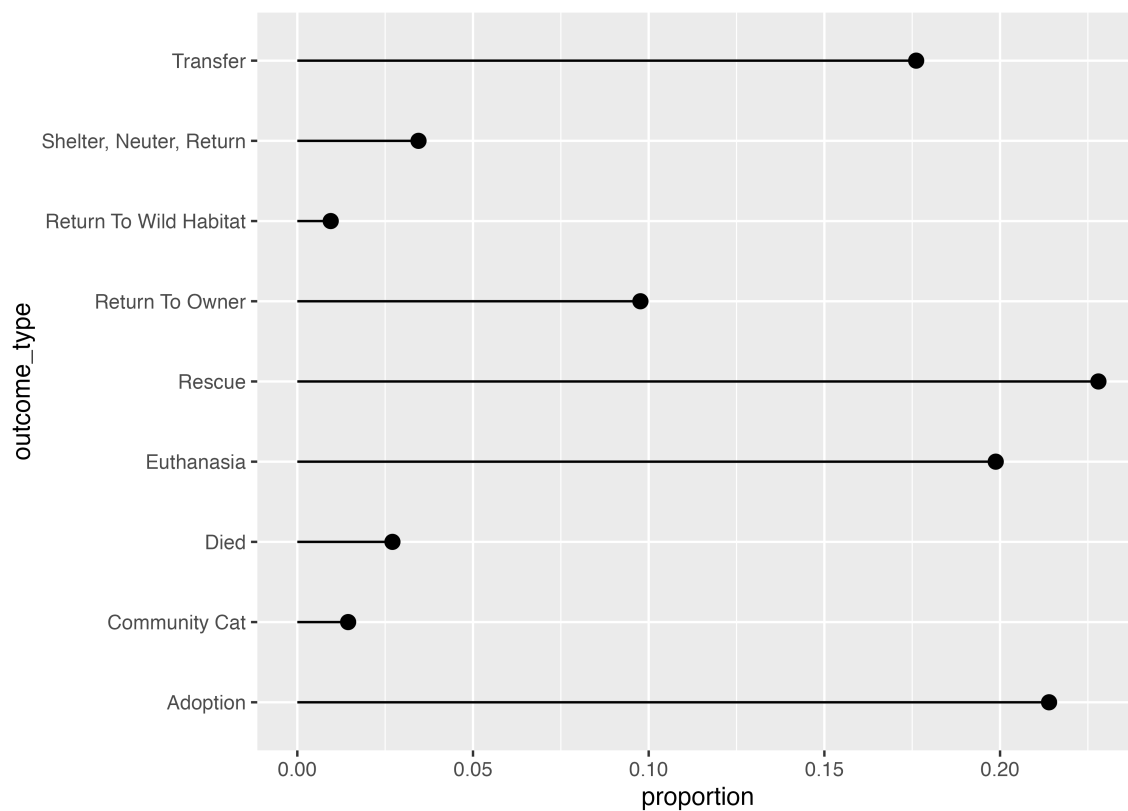
Use `group_by()` and `summarize()` and `mutate()` to create a data frame named `outcome_props` that looks like this (hint—that proportion column is calculated like `total / sum(total)`):

```
# A tibble: 9 × 3
  outcome_type      total proportion
  <chr>          <int>      <dbl>
1 Adoption       5651      0.214
2 Community Cat   382       0.0145
3 Died           715       0.0271
4 Euthanasia     5250      0.199
5 Rescue         6022      0.228
6 Return To Owner 2579      0.0976
7 Return To Wild Habitat 251      0.00950
8 Shelter, Neuter, Return 911      0.0345
9 Transfer       4652      0.176
```

```
# Do stuff here
```

Recreate lollipop plot

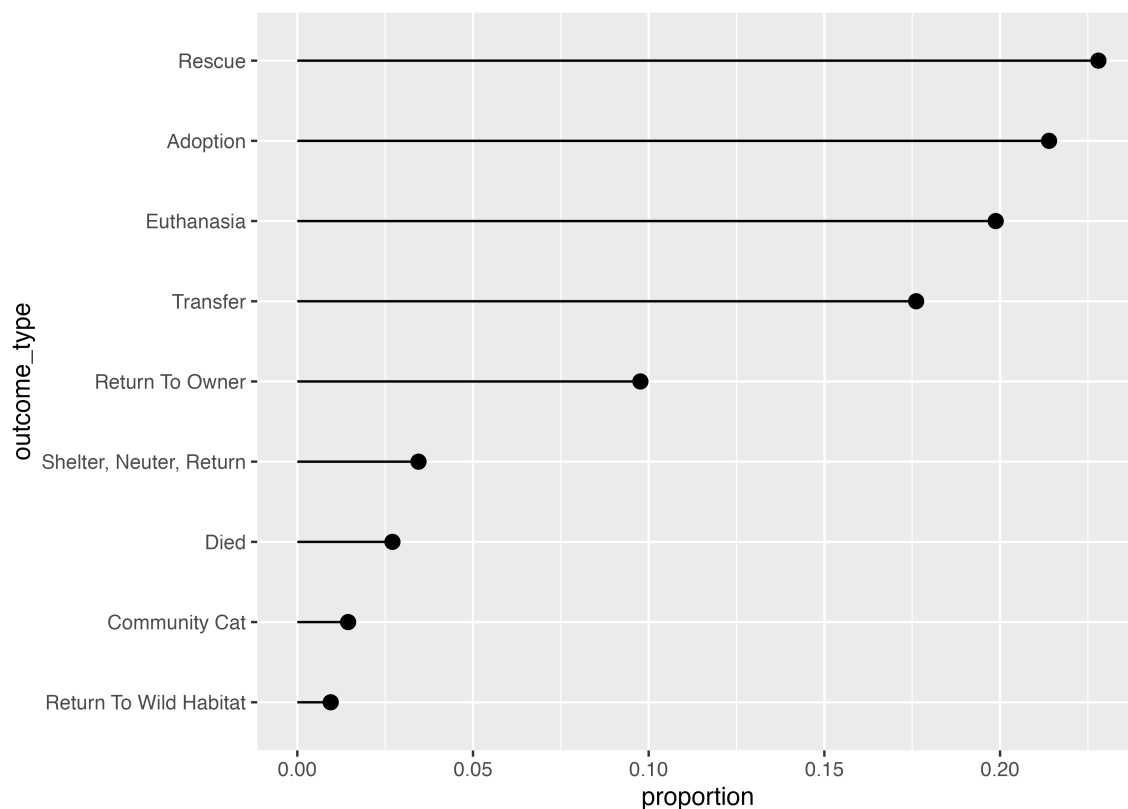
Run the code chunk below to see a plot.



Use `ggplot()` and `geom_pointrange()` in the chunk below to re-create the plot above using the `outcome_props` dataset.

```
# Do stuff here
```

Currently the outcome types are sorted alphabetically, which isn't super helpful. Sort them by proportion and recreate the plot below.



Big hint! Add this to the end of the code to make `outcome_props` to order `outcome_type` by the `proportion` column:

```
CODE_FOR_OUTCOME_PROPS |>
  mutate(outcome_type = fct_reorder(outcome_type, proportion))
```

```
# Do stuff here
```

Intake type over time

Are there seasonal patterns in the types of animal intake at the shelter?

Recreate data summary

Use `group_by()` and `summarize()` and `mutate()` to create a data frame named `month_intake_props` that looks like the output below.

You'll need to group by two variables to make the summary. IMPORTANTLY you'll also need to add another `group_by()` in between `summarize()` and `mutate()`, otherwise, R will calculate percentages in unexpected groups.

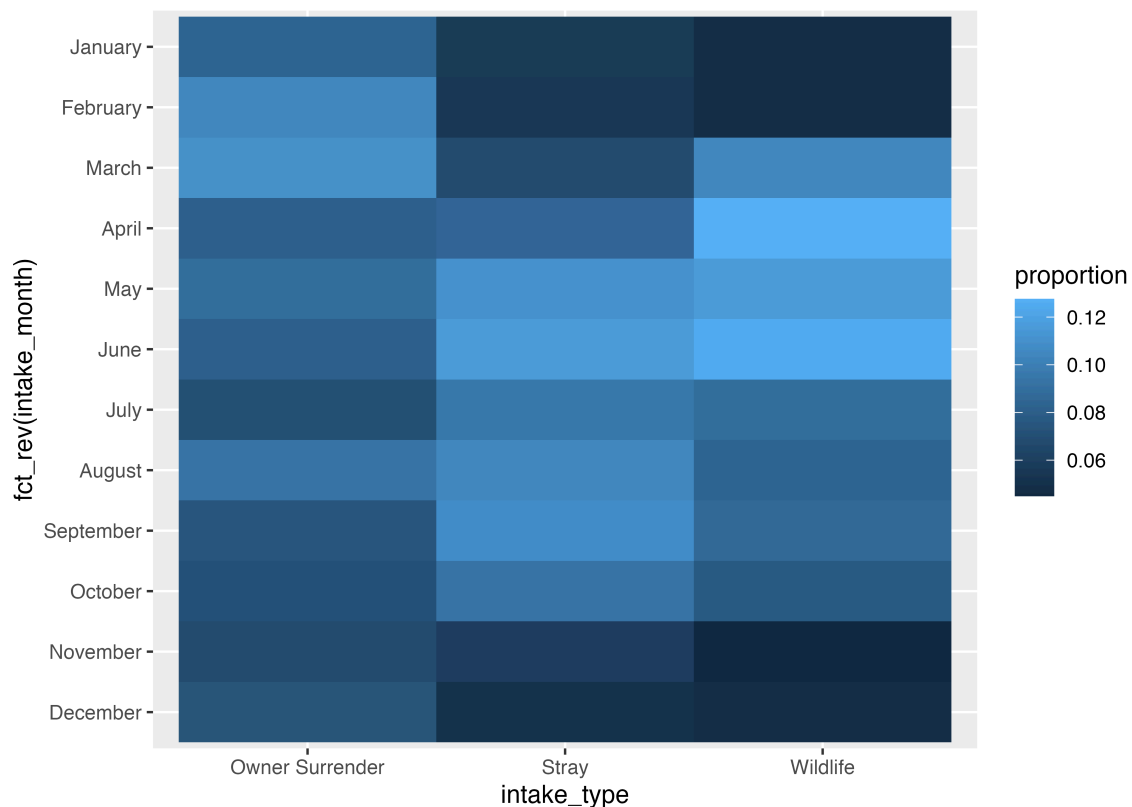
If you want the percentages of months to add up to 100% in each intake type (i.e. the percent of adoptions in January, February, March, etc.), you'll want to group by `intake_type` before calcu-

lating the proportion; if you want the percentages of intake types to add up to 100% in each month (i.e. the percent of all different intake types in January), you'll want to group by `intake_month` before calculating the proportion.

```
# A tibble: 36 × 4
# Groups:   intake_type [3]
  intake_type intake_month total proportion
  <chr>       <ord>      <int>     <dbl>
1 Owner Surrender January      196    0.0844
2 Owner Surrender February     240    0.103
3 Owner Surrender March        254    0.109
4 Owner Surrender April        189    0.0814
5 Owner Surrender May          206    0.0887
6 Owner Surrender June         190    0.0818
7 Owner Surrender July         165    0.0710
8 Owner Surrender August       214    0.0921
9 Owner Surrender September    175    0.0753
10 Owner Surrender October     167    0.0719
# i 26 more rows
# i Use `print(n = ...)` to see more rows
```

Recreate heatmap

Run the code chunk below to see a plot.



Use `ggplot()` and `geom_tile()` in the chunk below to re-create the plot above using the `outcome_props` dataset.

Hint: By default, the months will appear reversed, with January at the bottom and December at the top. Use `y = fct_rev(intake_month)` to reverse them.

```
# Do stuff here
```

Task 3: Extension

Copy the code for one of your plots above and paste it into the chunk below. Do some extra things to it, like changing the labels, changing the colors, adding a new geom, plotting a different variable from the dataset (there are lots of them!), changing the theme, or whatever. **This is your chance to play around and explore.**

```
# Do stuff here
```

Bibliography